Homework 2, DDL: April 17, 2026

Python Programming

1. Lab # 3 (4 Pts)
2. Lab # 4 (4 Pts)

Questions for Tech Report (6 Pts, $\leq$ 3 Pages)

1. Write down your derivation from (23) to (25), and derive the chain rule of $\nabla_h L$ in the case using the RBF kernel $K(\boldsymbol{x}_n, \boldsymbol{x}_{n'}) = \exp(-\|\boldsymbol{x}_n - \boldsymbol{x}_{n'}\|_2^2/h)$. (3 Pts)
2. **Sample selection for Kernel regression:** Given $\{y_n, \boldsymbol{x}_n\}_{n=1}^N$, we can learn a kernel regression model by

$$\min_{\boldsymbol{\alpha} \in \mathbb{R}^N} \|\boldsymbol{y} - \boldsymbol{K}\boldsymbol{\alpha}\|_2^2 + \lambda \mathcal{R}(\boldsymbol{\alpha}) \tag{34}$$

   Please design a regularizer $\mathcal{R}$ to encourage a sparse $\boldsymbol{\alpha}$, and provide an algorithm to obtain the optimal solution. (3 Pts)